
MLPC Report

Team Quantized Encoders

Samuel Eder

Andreas Resch

1 Annotation Verification & Qualitative Analysis

1.1 Summary of Disagreements

After carefully checking the assigned audio files, we have observed some trends in disagreements and violations.

1.1.1 Temporal Boundary Discrepancies

Annotators frequently disagreed on the on- and offset of events, such as stretching labels over non-target noise or failing to capture the full duration of the target sound (e.g. only the first bell tune was captured and not the full duration of the bell sound).

1.1.2 One-Second Rule Violations

It didn't happen as often as other rule violations or disagreements, but there were some cases where the one-second rule was clearly violated, labeling a long pause during two events of the same class as this class as well.

1.1.3 Class Misidentification

Very often we observed disagreement paired with rule violations when someone labeled target sounds as the wrong class, based on the scene description in the metadata. Very popular errors of this kind were misclassifying different door sounds as "door_open_close" even though they would have belonged to other classes, e.g. "microwave" or "wardrobe_drawer_open_close".

1.1.4 Omission of Audible Events

Annotators often missed clear target sounds described in the metadata, such as footsteps during a microwave sequence or at the beginning of a vacuuming recording.

1.2 Case Study

In this case study, we look at the file 005699.wav in detail and examine the mistakes that have been made and the disagreements that occurred. A visual representation of the annotations from HumanSignal can be found in Figure 1.¹

The biggest rule violations in these annotations are misclassifications of the microwave door, as discussed in section 1.1.3. Every single annotator managed to misclassify or not classify the microwave door handling. Looking at Figure 1, one can see three different annotations for the audio. In annotation 1 (annotation at the top), microwave door handling was not annotated at all. In annotations 2 (annotation in the middle) and 3 (annotation at the bottom), the door handling was annotated as "door_open_close" and as "wardrobe_drawer_open_close". Both are the violet markers in Figure 1 in the respective annotation.

To address disagreement between the annotators: annotation 3 has footsteps annotated during the microwave, which is the correct version. The footsteps are clearly audible, but annotations 1 and 2 do not have them annotated.

¹HumanSignal Website: <https://humansignal.com/>, Annotation Interface: <https://app.heartex.com/>

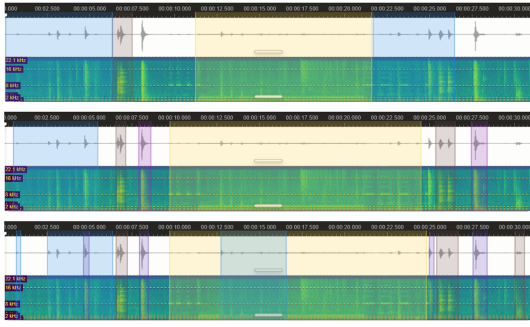


Figure 1: Side by side view of all annotators of audio file 005699.wav.

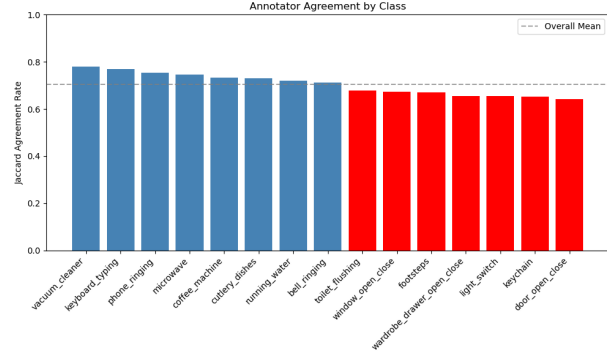


Figure 2: Annotator agreement by class.

2 Annotations

2.1 Annotator Agreement

We quantified the agreement using the method presented in the tutorial notebook, resulting in the overall Jaccard Agreement Rate. This agreement rate over all audio files from the development dataset is 71.0%.

Then we divided the development set into subsets, split by target classes. This means that every audio file that has e.g. "keyboard_typing" as one of the target classes, got included in the "keyboard_typing" subset. The mean Jaccard agreement rate was computed for every subset and stored in a dictionary. The results can be reviewed in Figure 2. The threshold is the overall mean, which is the mean of all mean Jaccard agreement rates and equals approximately 70.5%.

It is visible that all agreement rates are between 60 and 80%. Classes that usually produce louder and longer sounds tend to lie above the threshold, while classes that produce more subtle and shorter sounds tend to lie below the threshold. This could be due to the fact that on- and offset might be less ambiguous for longer sounds like "vacuum_cleaner" or "microwave", or just easier to annotate. Also, because these are typically longer sounds, the overlap between different annotators might be generally higher than for the shorter sound classes, e.g. "light_switch" or "keychain". Overall, 71% are an indicator of good overall agreement, however, there is still some ambiguity left.

2.2 Conversion of Annotations to Labels

2.2.1 Aggregation and Binarization

To convert the annotations to labels, we modified code from the tutorial notebook. The results were stored in a dictionary, using the file ID as key and the labels as values.

Before aggregation, the annotations were represented in the shape [T,C,A], where T are the time steps, C are the target classes, and A are the annotator. We aggregated across annotators by averaging over the annotator dimension. After averaging, the output shape was [T,C].

The binarization relies on a threshold. We decided to use the principle of majority vote and labeled "1", if the mean was greater than or equal to 0.5; everything below 0.5 was labeled "0".

2.2.2 Potential Imperfections

The binary mask, indicating if an annotator was also recording the scene is currently unused, as well as strategies to deal with potential tie breaks are not implemented. A combination of both, weighing the annotators own recordings slightly heavier could have been a way to deal with disagreements. Depending on the weight, this could change the outcome of binary disagreements. If there are three parties involved, it could still lead to a wrong result if the majority of the annotations indicate a different annotation than the creator.

2.3 Label Characteristics

There appears to be a relationship between device placement, taken from the metadata, and certain target classes, as illustrated in Figure 3. In general device placement is an indicator if the sound is dynamic, where the source of the

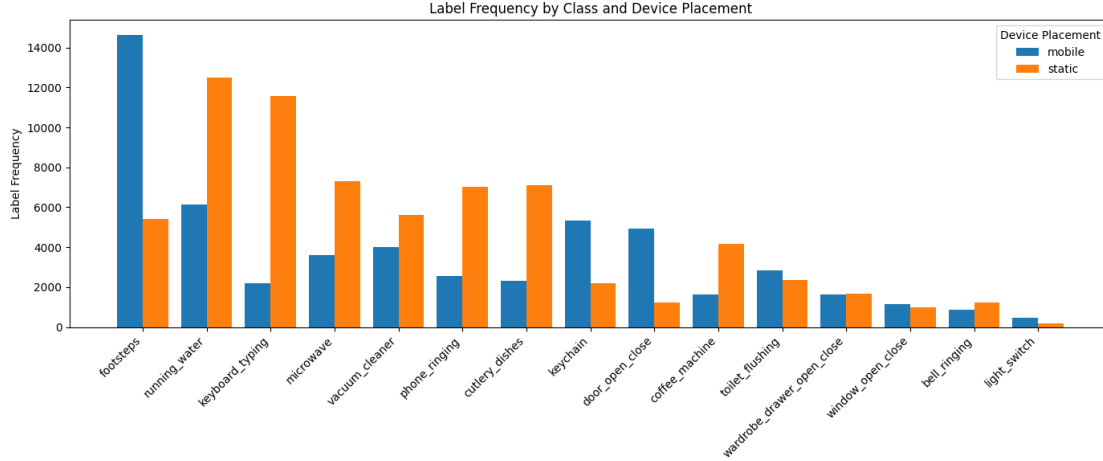


Figure 3: Label Frequency by Class and Device Placement

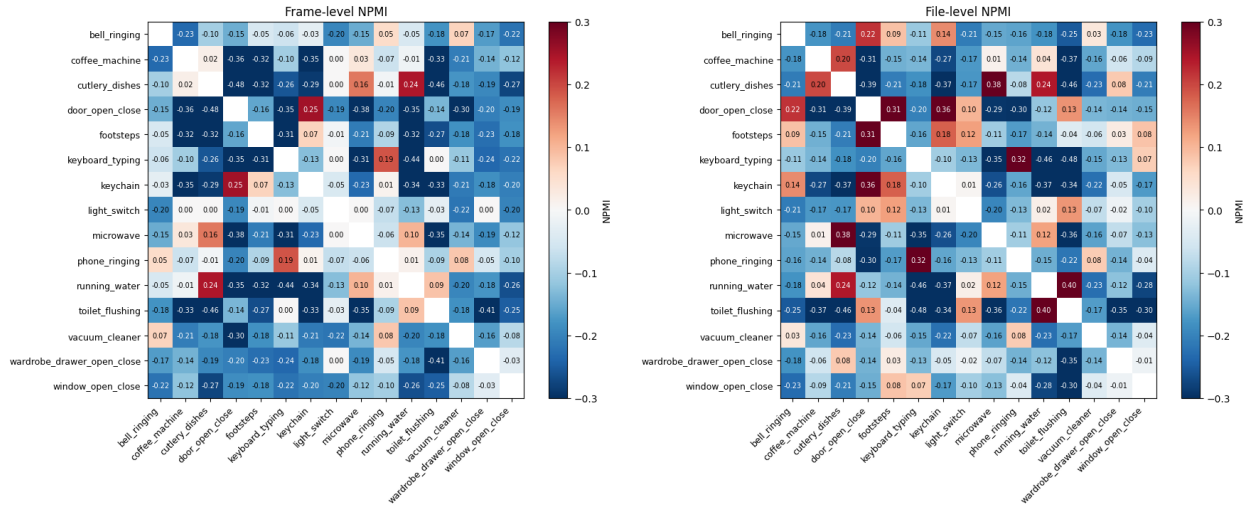


Figure 4: Co-occurrence Plots

sound might move as well, such as "footsteps" or "keychain", or static, like "running_water" or "keyboard_typing". To quantify the numbers, we simply counted each class occurrence in all audio files. The frequency is given in time steps.

The co-occurrence analysis reveals that most pairs have a low or negative score, indicating that overlapping sounds are less common, and most sounds are independent. as shown in Figure 4 on the left side, where the frame-level is shown, which explicitly requires overlapping sound events. The less conservative plot on the right side shows the file-level NPMI scores, which only require the sounds to occur in the same audio file. Therefore, pairs like "footsteps" and "door_open_close" can have a low score on frame-level, but a high score on audio-level, indicating that these might be sounds that do not occur at the same time naturally but still show a certain relationship within audio scenes.

3 Audio Features

3.1 Metadata

We examined the distribution by environment and the distribution by recording device placement, shown in Figure 5 and Figure 6. We restrict the analysis to the top 7 environments to reduce sparsity and improve interpretability. This indicates a class imbalance across environments, which may bias models toward dominant contexts such as kitchen recordings. "Mixed" means that there is more than one environment stated in the metadata. The device placement is

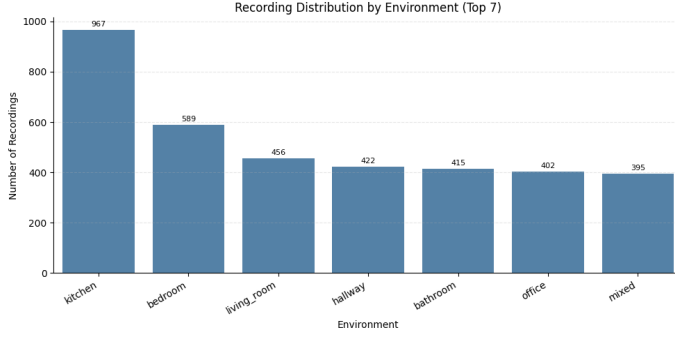


Figure 5: Recording Distribution by Environments, Top 7

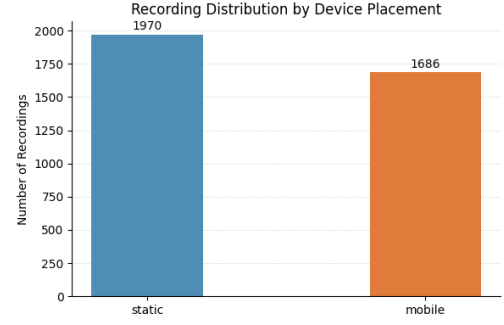


Figure 6: Recording Distribution by Device Placement

nearly balanced, showing 1970 recordings in static device placement and 1686 recordings in mobile device placement. Both placement types are well represented.

3.2 Feature Statistics

Table 1: Feature Statistics

Feature	Mean	Std	Range
ZCR	0.192	0.093	0.943
Energy	31.178	32.744	1978.073
Power	62.717	121.911	67940.875
Spectral Flux	1.929	1.703	166.599
Flatness	0.088	0.062	1.000
Centroid	2208.360	632.617	7615.978
Bandwidth	2785.426	377.676	4951.661
Rolloff Low	300.506	210.128	7437.500
Rolloff High	5074.285	960.228	7937.500
MFCC	-2.051	0.391	169.512
MFCC delta	0.000	0.000	32.648
MFCC delta2	-0.000	0.000	18.633
Mel Spectrogram	4.494	1.169	13.180
Contrast	15.737	0.928	80.107

For this task, the mean, standard deviation and range were computed for all features. This can be seen in Tabel 1. The features exhibit substantial variability. The range of the power feature is extremely high, which suggests strong variability in signal amplitude, potentially due to recording conditions and dynamic sound events. The statistics suggest a diverse dataset with strong variation in amplitude-related features and more stable behavior in structural spectral characteristics, such as "Spectral Flux", "Bandwidth", or rolloff features.

3.3 Feature Correlation

The feature correlation analysis was performed using Pearson correlation on frame-level averaged features. As shown in Figure 7, clear dependencies emerge: Spectral Flux and Energy are strongly correlated, linking signal amplitude to spectral variation, while zero-crossing rate and spectral centroid indicate that noisier signals tend to have higher spectral brightness. Correlations are also observed between MFCCs and Mel-spectrogram features. Although MFCCs are derived from Mel-scaled spectra, this likely reflects aggregated statistics rather than direct redundancy. Overall, energy-related and spectral features form correlated groups, suggesting potential redundancy relevant for feature selection.

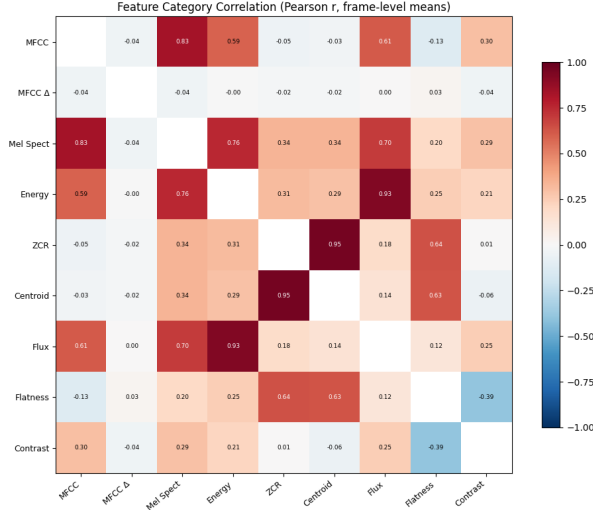


Figure 7: Feature category correlation (Pearson r) based on frame-level means.

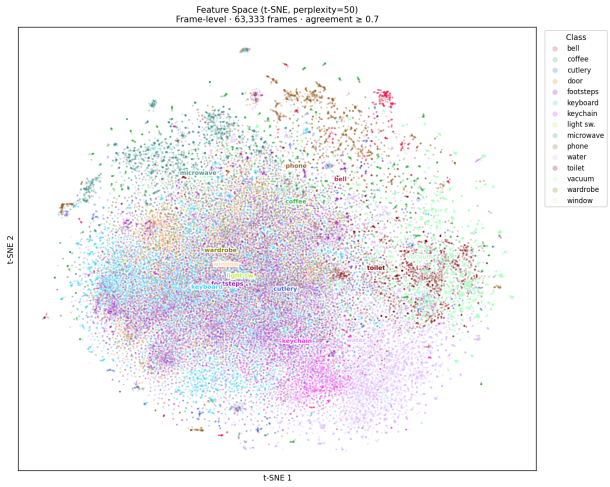


Figure 8: t-SNE visualization of the feature space

3.4 Feature Space Visualizations

To further analyze the feature space, we combined the correlation analysis (Figure 7) with a low-dimensional t-SNE visualization of frame-level feature vectors (Figure 8). While the correlation heatmap highlights strong dependencies between feature groups, the t-SNE projection shows a dense and largely overlapping distribution, indicating limited separability. Only a few classes (e.g., vacuum, toilet, microwave) form partially distinct regions, whereas short and transient events remain highly mixed. This suggests that the current frame-level representation captures insufficient discriminative structure, likely due to the lack of temporal context. Overall, incorporating longer temporal information may be necessary to improve separability.

4 Conclusion

4.1 Biases

The dataset contains several biases. There is a class imbalance, since some sound events occur significantly more frequently than others. Also, recording environments are not distributed uniformly. Then there are also annotator biases, shown in the varying agreement rates.

4.2 Conclusion

The dataset provides a solid foundation for the training stage, however, the biases discussed in section 4.1 can pose challenges for learning robust classifiers. Noisy labels and correlated features could lead to overfitting if not addressed properly.

4.3 Broader Impact

Public availability of a dataset like this can support faster development of many useful applications, such as smart homes or assistive systems. On the contrary, misuse for surveillance applications cannot be ruled out. Dataset creators might incorporate strict usage rules and censor sensitive information to minimize these risks.

5 Disclosure of LLM and AI Tool Use

Throughout the creation of this report, Large Language Models (LLMs) were used to support various stages of the workflow, including coding, analysis, explanations and writing. A mentionable exception is section 1, which was completed entirely without the use of LLMs. Also section 2.3 was the only part of section 2 that involved the use of LLMs.

A more detailed list of AI Use:

- Claude Code:
 - Coding assistance to generate the plots in a jupyter notebook.
A list of plots that used the help Claude Code:
 - * Figure 4
 - * Figure 7
 - * Figure 8
 - * Table 1
 - explanations and analysis of different audio features for deeper understanding
- ChatGPT:
 - Explanation of LaTeX related questions regarding formatting issues and error messages
 - Assistance in explanations of task descriptions
 - Explanation of technical questions regarding the dataset
 - Refining the written paragraphs and textual interpretations
 - Shortening the text to fit the word limit